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Practice questions

1) Broken telephone

Consider the following simplified version of the broken telephone game. A first player secretly communicates a message $m_1 \in \{0, 1, 2\}$ to a second player with probabilities

$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \frac{1}{3}.$$

The second player then secretly communicates a message $m_2 = m_1 + n$ to a third (final) player, where

$$P(n = 0) = \frac{2}{3}, \quad P(n = +1) = \frac{1}{3}.$$

- (a) What is the optimal Bayes classification of the message received by the last player?
- (b) What is the expected error probability?

2) Binary classification

Consider the following binary classification problem with losses $L(C, C_{\text{pred}})$:

	$C_{\text{pred}} = +1$	$C_{\text{pred}} = -1$
$C = +1$	0	100
$C = -1$	10	0

Your prior is

$$P(C = +1) = \frac{1}{10}.$$

- (a) What is the optimal classifier (and expected loss) based only on the prior distribution?
- (b) Suppose you can run an experiment with binary outcomes $x \in \{0, 1\}$ with likelihoods

$$P(x = 1 \mid C = +1) = \frac{3}{4}, \quad P(x = 1 \mid C = -1) = \frac{1}{20}.$$

Find the classifier that minimizes expected loss. Compute the optimal expected loss and compare with part (a).